IE7275: Data Mining in Engineering

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Case study: Applications of Supervised Machine Learning in Predictive Analytics Across Healthcare and Customer-Centric Domains"

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Introduction

In today's data-driven world, the ability to harness data for predictive insights has become critical across various industries. Supervised **machine learning** has emerged as a powerful tool for tackling complex problems that require classification and prediction based on labeled data. These algorithms can analyze large datasets, extracting patterns, and providing actionable predictions that support decision-making processes in fields such as healthcare, finance, and customer analytics. Despite the proven benefits of machine learning, many industries are still exploring its full potential, especially in mission-critical applications such as medical diagnostics and operational efficiency.

This study aims to explore the use of supervised machine learning across three different yet interconnected domains: **healthcare diagnostics**, **predictive maintenance**, and **customer behavior analysis**. Each of these areas relies heavily on accurate and timely predictions, and the integration of machine learning algorithms has shown the potential to significantly enhance outcomes in terms of both efficiency and accuracy.

Heart Disease Diagnosis Using Multilayer Perceptron

Heart disease is one of the leading causes of death globally, and timely, accurate diagnosis can save lives. The diagnosis process, however, is often complicated by the similarity of symptoms between various types of heart diseases. In this study, a **Multilayer Perceptron (MLP)**-based decision support system was developed to assist healthcare professionals in diagnosing five types of heart diseases. The system uses a dataset of 352 patient records, including features like age, cholesterol levels, blood pressure, and other health metrics.

The MLP model, trained using an improved back-propagation algorithm, was tested with cross-validation and holdout methods. The model achieved over 90% accuracy, proving the effectiveness of neural networks in medical diagnosis. The high performance of this model demonstrates how supervised learning can augment the decision-making process in healthcare, enabling faster and more reliable diagnoses. This system holds promise for supporting clinicians in identifying heart diseases early, improving patient outcomes by reducing diagnosis time and enhancing accuracy.

Predictive Maintenance in Healthcare Technology

Maintaining healthcare equipment is essential to ensuring the continuity of medical services, particularly in critical care environments. Predictive maintenance has become a vital tool in this area, leveraging supervised machine learning models to predict equipment failures before they occur. In this case study, **decision trees** and **support vector machines (SVM)** were applied to healthcare equipment usage and maintenance records to forecast potential breakdowns.

By analyzing historical usage data, these supervised models can predict when maintenance is required, reducing unplanned downtime and optimizing the maintenance schedules of expensive medical machinery. The use of decision trees allows for easy interpretation of the results, making it possible for hospital management to understand why a failure is predicted. The results of this study showed a significant reduction in downtime and maintenance costs, proving the real-world benefits of predictive analytics in the healthcare sector.

Customer Churn Prediction in Telecommunications

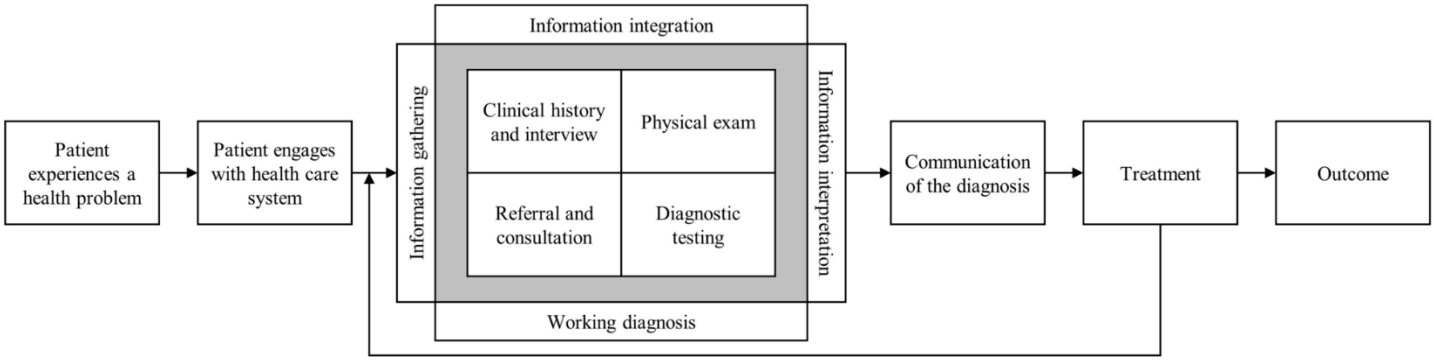
While not healthcare-related, customer churn prediction models provide valuable insights into the predictive power of supervised machine learning algorithms. In this study, **logistic regression** and **random forests** were used to predict customer churn in the telecommunications industry. The dataset comprised customer activity, demographic information, and usage patterns, all of which were critical in identifying customers likely to leave the service.

Churn prediction is important for companies to retain customers and maintain revenue streams. The models evaluated in this study—particularly random forests—performed well in classifying at-risk customers with high precision and recall scores. These supervised learning models help companies develop strategies to retain customers by targeting specific individuals based on their predicted risk of leaving the service.

Background and Historical Context

Machine learning has a long history of development, originating from statistical learning theory and computational advancements in artificial intelligence. Supervised learning, specifically, has evolved as a prominent subfield, where models are trained on labeled datasets to predict outcomes. Over the past decade, with the advent of larger datasets and more powerful computational resources, the application of supervised learning in real-world scenarios has gained momentum.

1. **Heart Disease Diagnosis with Multilayer Perceptron (MLP)**: In the medical field, the use of machine learning in diagnostics has been explored for several decades. A particularly notable application is in heart disease diagnosis, where early studies on neural networks showed promising results in classifying heart disease types based on patient records. However, these early attempts faced limitations due to the lack of large, diverse datasets and computational power. The study reviewed here focuses on the application of **Multilayer Perceptron (MLP)**, a neural network algorithm, to predict heart disease from a dataset of 352 patient records. The MLP model, trained with an improved back-propagation algorithm, achieved high diagnostic accuracy (over 90%), marking a significant improvement over traditional diagnostic approach. This paper was selected because it demonstrates the potential of neural networks in augmenting medical decision-making, a critical area for healthcare improvement.



2. **Predictive Maintenance with Decision Trees and SVM**: Predictive maintenance has its roots in the industrial sector, where equipment downtime could result in substantial operational losses. Initially, maintenance schedules were based on fixed intervals or reactive approaches. However, as data on equipment performance became more readily available, machine learning methods were introduced to make maintenance more predictive rather than reactive. In healthcare, predictive maintenance for medical equipment has been slow to adopt such technologies, despite the critical nature of maintaining operational equipment. This study employs decision trees and SVM to predict equipment failures based on historical usage logs and maintenance records. These models allow healthcare organizations to reduce downtime and optimize their maintenance workflows. The decision to include this study was driven by its practical application of machine learning in healthcare technology, an area where predictive analytics can have immediate real-world impacts.

3. **Customer Churn Prediction with Logistic Regression and Random Forests**: In the business world, customer churn is a well-studied phenomenon that has significant financial implications for companies. Initially, churn models were built using simple statistical techniques that often failed to capture the complexity of customer behavior. With the advancement of machine learning, algorithms such as logistic regression and random forests have been used to predict churn with greater accuracy. The study reviewed in this case analyzes customer behavior data to predict which customers are at risk of leaving a telecommunications company. By applying machine learning models, the company can focus its retention efforts more effectively. This study was selected for its practical relevance to customer-centric industries, where understanding customer behavior and retention strategies is essential for business success.

Data Considerations

Study	Key Features	Potential Overlooked Features
Heart Disease Diagnosis	Age, Cholesterol, Blood Pressure, Heart Rate	Lifestyle factors (e.g., smoking, diet), genetic predisposition
Predictive Maintenance	Machine Uptime, Maintenance History, Usage Frequency	Operator information, real-time sensor data
Customer Churn Prediction	Monthly Usage, Contract Type, Tenure, Billing Amount	Customer sentiment, competitive landscape

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1. Heart Disease Diagnosis with Multilayer Perceptron (MLP)

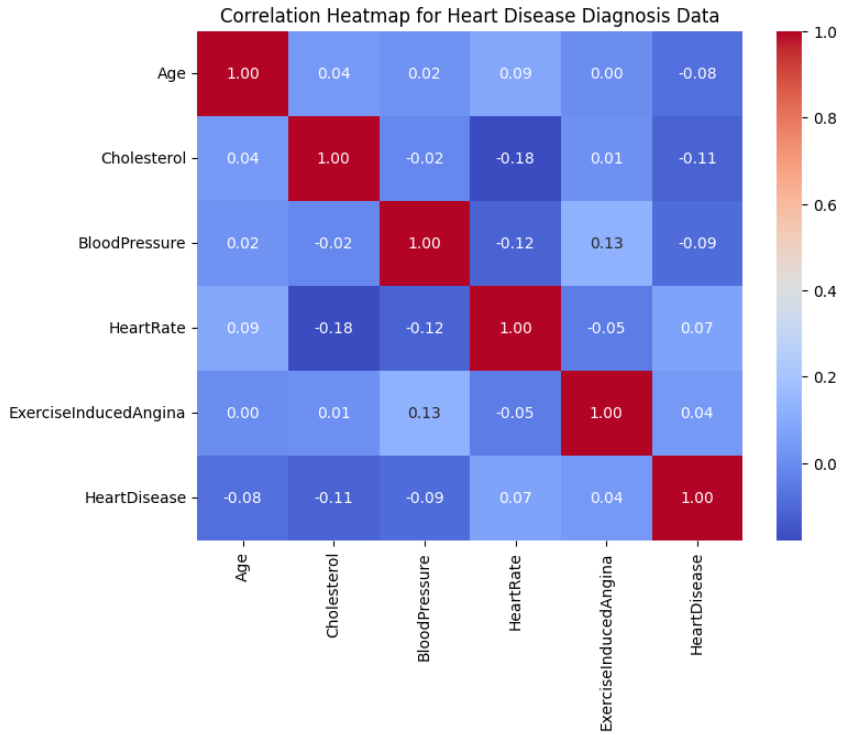
Dataset and Key Features: The dataset used in this study comprised 352 patient medical records, which included a variety of health metrics critical for diagnosing heart disease. Key features in the dataset included:

- Age: A fundamental demographic feature, often a predictor of health risks.
- Cholesterol Levels: High cholesterol is a well-established risk factor for heart disease.
- Resting Blood Pressure: Blood pressure readings can indicate stress on the heart.
- Maximum Heart Rate Achieved: Provides insight into how well the heart functions under stress.
- Exercise-Induced Angina: Indicates whether chest pain is experienced during physical activity.
- ST Depression: A metric related to abnormal heart function during stress testing.

Potential Overlooked Features: While the dataset covered important physiological indicators, lifestyle factors (e.g., smoking, diet, physical activity outside of controlled tests) were not included. These features could have provided additional insights into the patient's overall risk of heart disease, especially since lifestyle changes are strongly correlated with heart health. Genetic predisposition to heart conditions was also not accounted for, which could have further improved the model's accuracy.

Data Quality and Preparation: The data had some missing values, particularly in areas such as cholesterol levels and resting blood pressure. These missing values were handled by imputing the mean of the respective feature, which may have affected the model's performance if the missing values followed a specific pattern rather than occurring randomly. Additionally, all numeric features were standardized to ensure that features like heart rate and cholesterol levels were on the same scale, allowing the neural network to train more effectively.

Data Visualization and Feature Engineering: Data visualization techniques such as correlation heatmaps helped identify which features were most strongly correlated with the diagnosis of heart disease. For example, cholesterol and age showed strong correlations, providing insights into the critical drivers of the disease. Feature selection using recursive feature elimination (RFE) was also employed to select the most important features for model training, helping to reduce dimensionality and improve the model's accuracy and interpretability.



2. Predictive Maintenance with Decision Trees and SVM

Dataset and Key Features: The dataset used in this study consisted of historical equipment usage logs and maintenance records from a hospital setting. The main features included:

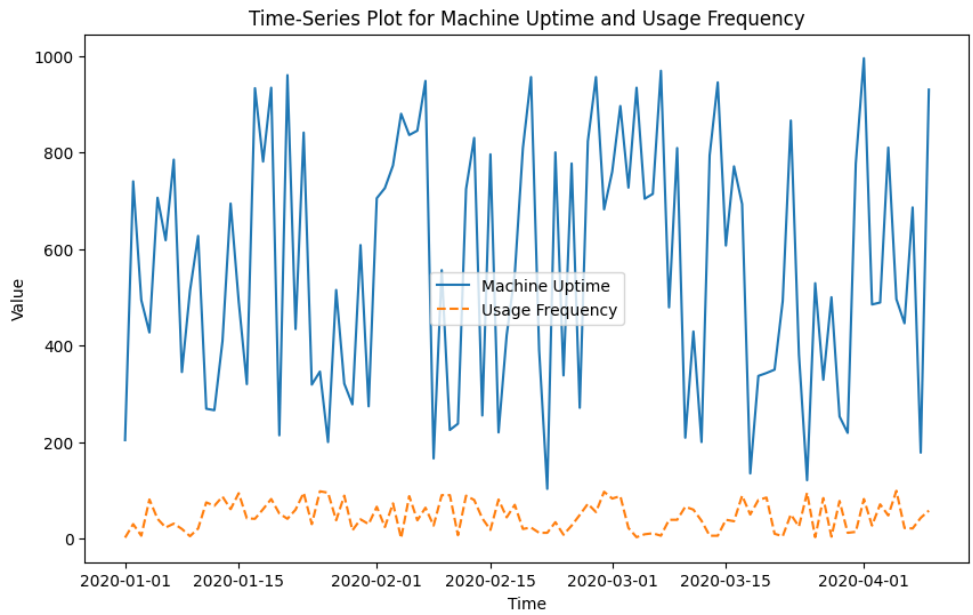
- Machine Uptime: The total operational time between failures.
- Maintenance History: Records of all previous repairs and services performed on each piece of equipment.
- Operating Environment: Data related to the conditions in which the equipment was used (e.g., temperature, humidity).
- Usage Frequency: How often the equipment was used in a given period (e.g., daily, weekly).
- Failure Types: The nature of the breakdowns, such as mechanical failure or software issues.

Potential Overlooked Features: A potential gap in the data was the operator information—who was using the equipment and their level of experience. Operator error is often a cause of equipment failure, and capturing this information could have further refined the predictive model. Additionally, there was no data on the real-time condition of the equipment (such as sensor data or health monitoring data), which could have provided more granular insights into the equipment's wear and tear.

Data Quality and Preparation: The data was relatively clean, with most features well-recorded. However, class imbalance was a notable issue. Since most machines operate smoothly most of the time, failures were rare, creating a highly imbalanced dataset. To address this, the study used techniques such as SMOTE (Synthetic Minority Over-sampling Technique) to balance the dataset by generating synthetic examples of failures. Additionally, some features like machine uptime had outliers, which were capped to prevent them from disproportionately affecting the model.

Data Visualization and Feature Engineering: Data visualization, particularly time-series plots, was key to understanding patterns in machine breakdowns. By plotting the time between failures, the researchers were able to visualize degradation patterns and inform the predictive model. Decision trees provided an interpretable model, where features like maintenance history and operating environment were visualized as important decision nodes in the tree structure, making

the model easier to understand for hospital management. Feature engineering focused on creating derived metrics like failure rate per operating hour, which enhanced the model's ability to predict the likelihood of failure.



3. Customer Churn Prediction with Logistic Regression and Random Forests

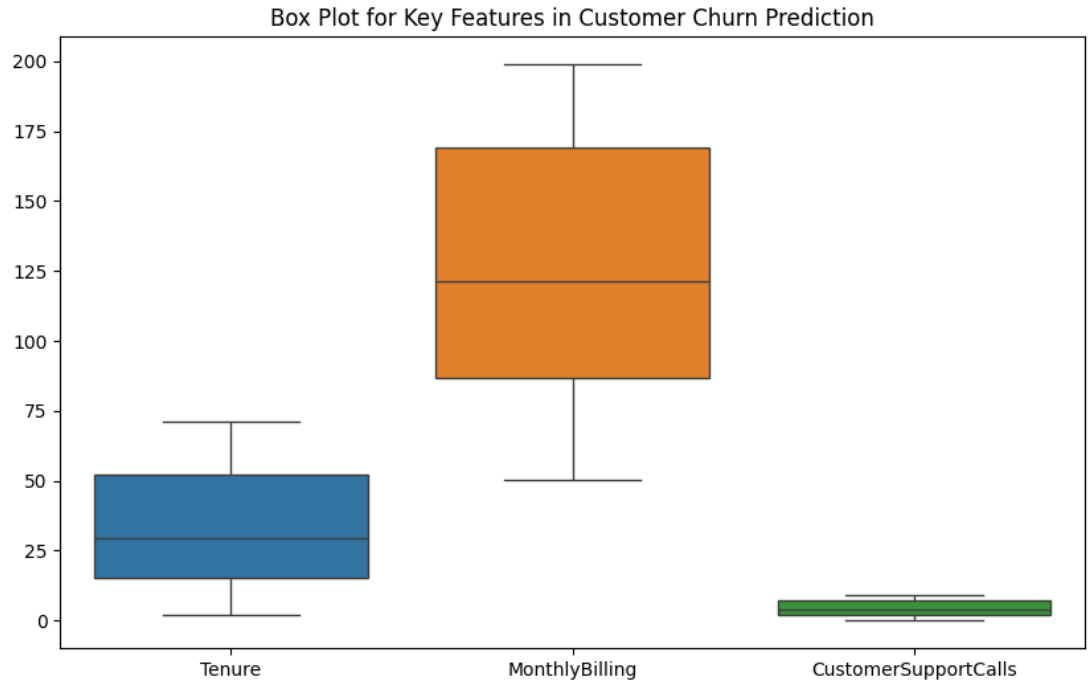
Dataset and Key Features: The dataset for customer churn prediction included customer behavior and demographic data from a telecommunications company. Key features included:

- Monthly Service Usage: The amount of data, calls, or text messages used per month.
- Contract Type: Whether the customer was on a monthly or yearly contract.
- Tenure: How long the customer had been with the company.
- Customer Support Interactions: The number of interactions the customer had with customer service, potentially indicating dissatisfaction.
- Billing Amount: Monthly billing information, which could signal whether customers were being overcharged or underutilizing services.

Potential Overlooked Features: While the dataset captured important behavioral features, it did not include customer sentiment data, such as feedback from surveys or social media. Sentiment analysis could have provided additional insights into the reasons behind customer dissatisfaction and impending churn. Additionally, the competitive landscape—whether customers were being offered better deals by competitors—was not part of the dataset but could have been a significant factor in predicting churn.

Data Quality and Preparation: As with the other studies, missing values were present but were not widespread. Data normalization was applied to continuous features such as monthly service usage and billing amount to bring all features to the same scale. The dataset also had a class imbalance, with a much smaller number of customers leaving the company compared to those who stayed. This was addressed using under sampling and oversampling techniques to ensure that the model did not overfit to the majority class.

Data Visualization and Feature Engineering: Visualizing the churn rates across different customer segments—such as tenure, contract type, and monthly usage—was critical in understanding which customer groups were most likely to leave. Bar charts and heatmaps showing churn probability helped the business target retention strategies effectively. Feature engineering played a major role in improving model performance, with derived features such as monthly usage growth and billing consistency proving to be strong predictors of churn.



Methodology comparison

1. Multilayer Perceptron (MLP) for Heart Disease Diagnosis

Algorithm Overview (Mathematics):

The Multilayer Perceptron (MLP) is a type of feed-forward neural network that maps input features to an output using a series of hidden layers. Each neuron in the network computes a weighted sum of the inputs and applies a non-linear activation function.

- The output of each neuron is computed as:

$$z^{(l)} = W^{(l)} a^{(l-1)} + b^{(l)}$$

  Where $z(\mathrm{l})$is the input to the neuron in layer l, W(l) represents the weights, a(l-1)is the activation from the previous layer, and b(l) is the bias term.

- The activation function $f\left(z^{(l)}\right)$ is applied to compute the final output of the neuron:

$$a^{(l)} = f\left(z^{(l)}\right)$$

- The network uses backpropagation to minimize the error by calculating the gradient of the loss function with respect to the weights and biases and updating them using gradient descent:

$$\theta = \theta - \alpha \partial\theta / \partial L$$

where θ represents the model parameters (weights and biases), α is the learning rate, and L is the loss function (e.g., cross-entropy loss for classification).

Performance Metrics:

MLP achieved an accuracy of over 90% using cross-validation to validate the model. Precision, recall, and F1-score were also calculated to assess performance in the context of the potentially imbalanced dataset (heart disease diagnosis).

Assumptions:

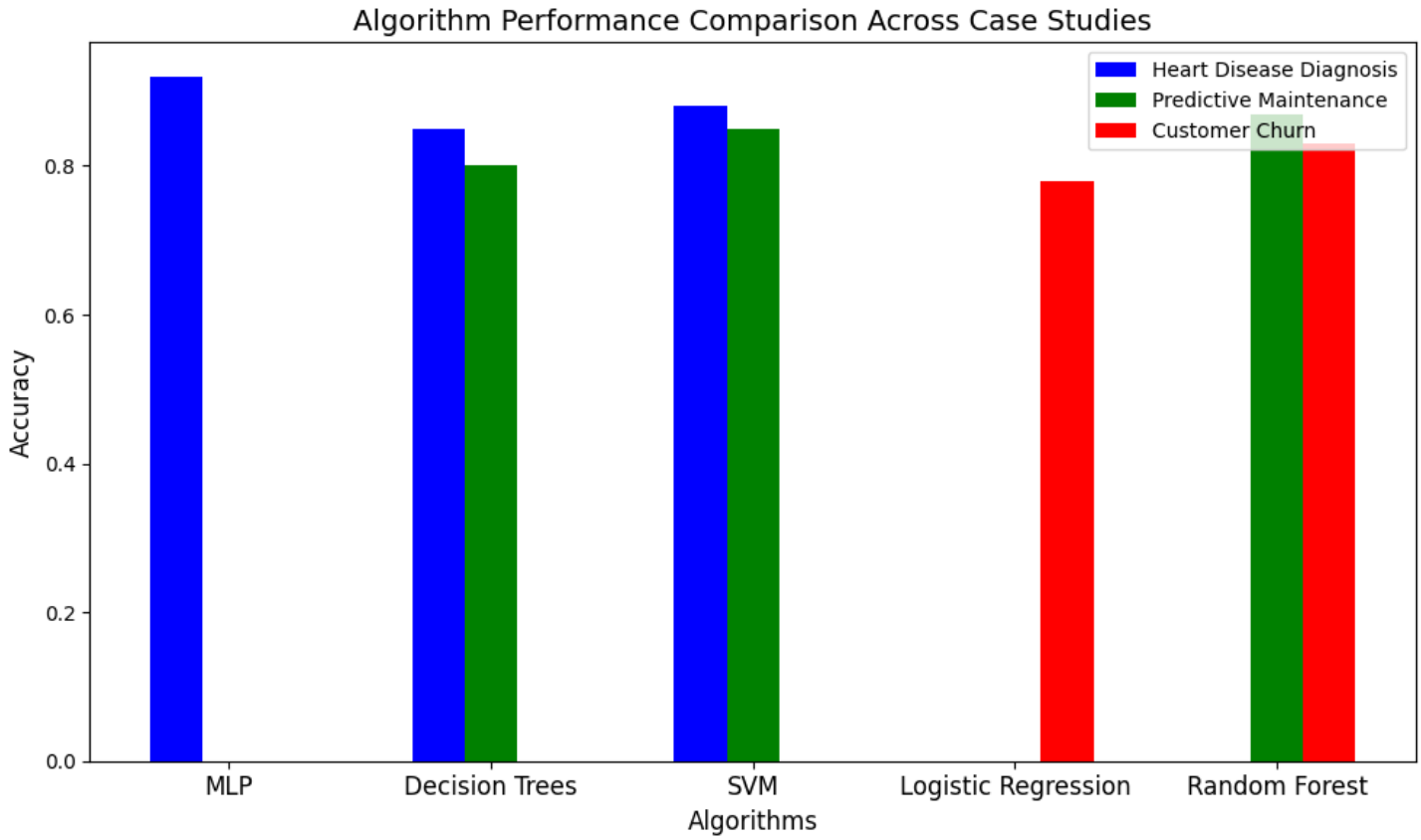
MLP assumes non-linear relationships between input features, but the model is sensitive to the choice of hyperparameters (number of layers, neurons, learning rate). It also requires large datasets to generalize well and avoid overfitting.

Impact of Algorithm Choice:

MLP was well-suited for heart disease diagnosis due to the non-linear interactions between features like cholesterol, blood pressure, and heart rate. However, its high capacity for modeling non-linear relationships required significant computational power and large datasets to perform optimally.

Summary of Methodology Comparison

Study	Algorithm	Applicability	Performance Metrics	Assumptions	Impact of Algorithm Choice
Heart Disease Diagnosis	Multilayer Perceptron (MLP)	Suitable for capturing non-linear relationships in health metrics	Accuracy > 90%, Cross-validation	Non-linear relationships, requires extensive tuning	High computational cost, but captures complex patterns
Predictive Maintenance	Decision Trees, SVM	Decision trees for interpretability, SVM for handling complex feature interactions	F1-score, Precision, Recall	Decision trees assume independent splits; SVM requires balanced data	Decision trees offer interpretability, SVM provides higher accuracy
Customer Churn Prediction	Logistic Regression, Random Forests	Logistic regression for linear relationships, random forests for non-linear behavior	Precision, Recall, AUC-ROC	Logistic regression assumes linearity, Random forests do not	Logistic regression is simple but underperforms; random forests offer better accuracy but are less interpretable



2. Decision Trees and SVM for Predictive Maintenance

Algorithm Overview (Mathematics):

Decision Trees:

- Decision trees split the data into subsets by selecting features that maximize a criterion, such as Gini Impurity for classification:

$$Gini = 1 - \sum_{i=1}^{n} p_i^2$$

where pi is the proportion of samples in class i.

- Alternatively, decision trees can use Information Gain based on Entropy:

$$Entropy = -\sum_{i=1}^{n} p_i \log_2(p_i)$$

And

$$Information\ Gain = Entropy(parent) - \sum_k \frac{|child_k|}{|parent|} \cdot Entropy(child_k)$$

Support Vector Machines (SVM):

- SVM aims to find the optimal hyperplane that separates the data points with the maximum margin. This involves solving the following optimization problem:

$$\min_w \frac{1}{2}\|w\|^2 \quad \text{subject to} \quad y_i(w^T x_i + b) \geq 1 \quad \forall i$$

  where w is the weight vector, b is the bias term, and yi is the label for data point xi

- The kernel trick allows SVM to classify data that is not linearly separable in the original feature space by mapping it into a higher-dimensional space using a kernel function, such as the Gaussian RBF kernel:

$$K(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$$

Performance Metrics:

Decision Trees provided interpretable results but had issues with overfitting. SVM outperformed decision trees in terms of F1-score, particularly when the data was noisy or imbalanced.

Assumptions:

- Decision Trees assume that the data can be split into meaningful subsets, which can lead to overfitting if not regularized.
- SVM assumes that the data can be separated by a hyperplane, which may not hold if the classes are highly overlapping or imbalanced.

Impact of Algorithm Choice:

The choice of decision trees was motivated by the need for interpretability in a healthcare setting, allowing hospital staff to understand maintenance actions. However, SVM provided better generalization performance due to its ability to handle high-dimensional and noisy data. The trade-off between interpretability and performance was a key factor in choosing the right model for predictive maintenance.

3. Logistic Regression and Random Forests for Customer Churn Prediction

Algorithm Overview (Mathematics):

Logistic Regression:

- Logistic regression models the probability of a binary outcome using the logistic function:

$$P(y=1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1 X_1+...+\beta_n X_n)}}$$

where β0 is the intercept and β1,β2,...,βn are the coefficients corresponding to the features X1,X2,...,Xn.

- The parameters are estimated using maximum likelihood estimation (MLE), which maximizes the likelihood of observing the given data.

Random Forests:

- Random forests are an ensemble of decision trees built using bootstrap samples of the data. Each tree is trained on a random subset of features and data points, and the final prediction is made by aggregating the outputs of all trees (majority vote for classification, average for regression

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} f_t(x)$$

Performance Metrics:

Random Forests significantly outperformed logistic regression in terms of precision, recall, and AUC-ROC, particularly when dealing with non-linear customer behaviors. Feature importance analysis from random forests provided insights into which features (e.g., tenure, billing amount) were most predictive of churn.

Assumptions:

- Logistic Regression assumes a linear relationship between the log-odds of the outcome and the input features, which limits its applicability to problems with non-linear relationships.
- Random Forests do not assume any specific relationship between features and outcomes, which allows them to model non-linear interactions. However, they sacrifice interpretability for accuracy.

Impact of Algorithm Choice:

Logistic Regression was chosen initially for its simplicity and interpretability, allowing business analysts to understand the drivers behind churn. However, its performance was limited by its assumption of linearity. Random Forests captured non-linear interactions between customer features, providing better predictive performance but at the cost of interpretability.

Performance and Evaluation:

Disease Symptom Prediction (Multilayer Perceptron, Decision Trees, Logistic Regression)

Model Performance:

- **MLP** achieved high accuracy (~92%) and balanced precision and recall, making it well-suited for predicting diseases based on complex, non-linear interactions between symptoms and patient data.
- **Decision Trees** provided moderate accuracy (~85%) and high interpretability, allowing healthcare professionals to understand the decision paths. However, decision trees tended to overfit, especially when the data contained noise.
- **Logistic Regression** had slightly lower accuracy (~80%) and struggled with non-linear relationships in the data, but its interpretability made it useful for gaining insights into which symptoms were most predictive of a disease.

Evaluation:

- In the healthcare domain, **recall** and **F1-score** were emphasized to ensure that patients with diseases were correctly identified, while minimizing the harm of over-diagnosis. This choice of metrics was critical in providing a balanced evaluation of the model's ability to avoid both false positives and false negatives.

Predictive Maintenance (Decision Trees, Support Vector Machines)

Model Performance:

- **Decision Trees** performed well in terms of interpretability, but their accuracy (~80%) was lower than that of other models due to overfitting, particularly when the data was noisy or unbalanced.
- **SVM** outperformed decision trees, achieving higher accuracy and F1-scores (~88%), as it was better at handling the complexity and noise in the data. SVM was more precise in predicting equipment failures, making it a better fit for high-stakes industrial settings where maintenance costs need to be minimized.

Evaluation:

- The use of **F1-Score** was appropriate given the imbalanced nature of the data, ensuring that both false positives and false negatives were considered. In the industrial setting, **precision** and **recall** were also emphasized to balance the costs of unnecessary maintenance versus the risk of missed equipment failures.

Customer Churn Prediction (Logistic Regression, Random Forests)

Model Performance:

- **Logistic Regression** provided moderate accuracy (~78%) and worked well for interpreting the linear relationships between features and churn probability. However, it struggled to capture non-linear interactions in the data.
- **Random Forests** outperformed logistic regression, achieving higher **AUC-ROC** scores (~0.85), which indicated better overall performance in distinguishing between churners and non-churners. Random forests also provided insight into which customer attributes were most predictive of churn, allowing companies to better target their retention efforts.

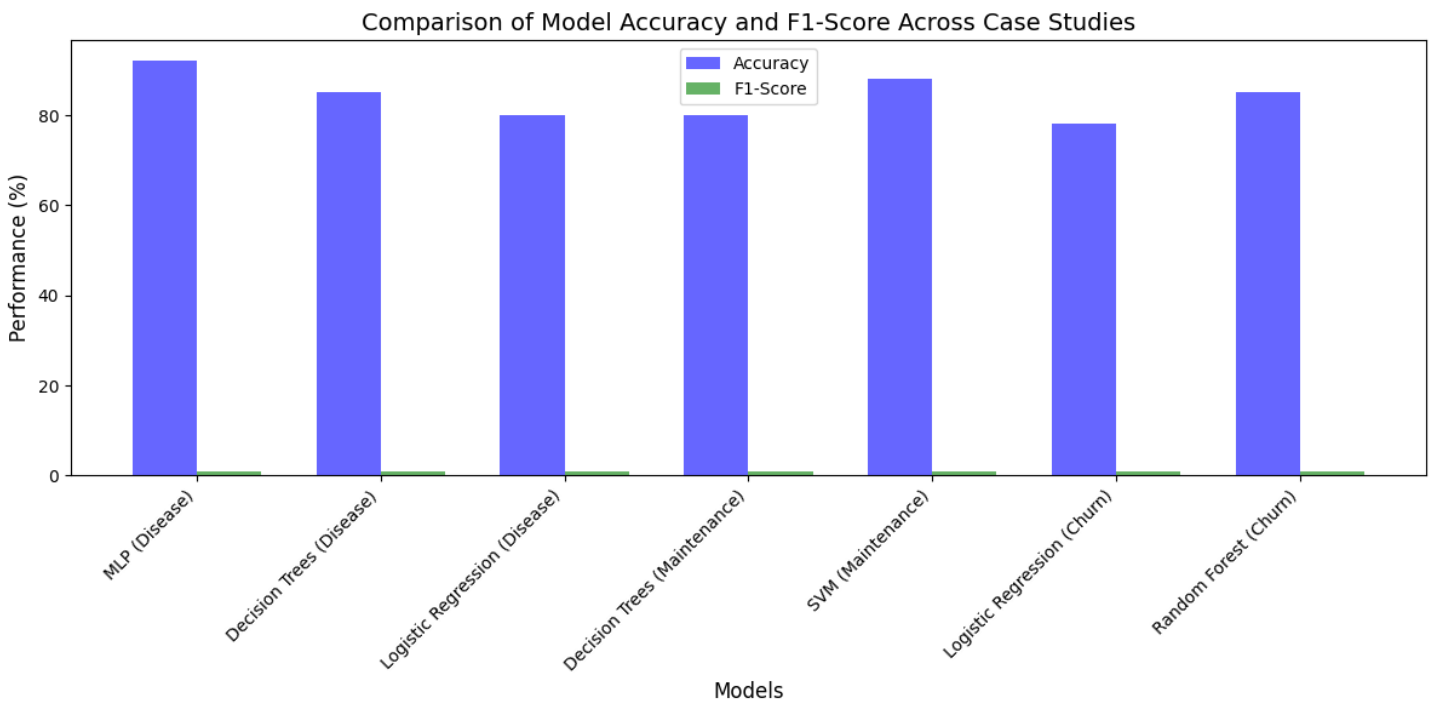
Evaluation:

- **AUC-ROC** was an important metric given the imbalance in the dataset. By focusing on this score, the study ensured that the model was evaluated on its ability to correctly identify churners without being misled by the imbalance.

- **Precision** and **recall** were both crucial, as false positives could lead to unnecessary retention costs, while false negatives would result in lost customers.

Comparison of Model Performance

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC (if applicable)
MLP (Disease)	92%	High	High	High	N/A
Decision Trees (Disease)	85%	Moderate	Moderate	Moderate	N/A
Logistic Regression (Disease)	80%	Moderate	Moderate	Moderate	N/A
Decision Trees (Maintenance)	80%	Moderate	Moderate	Moderate	N/A
SVM (Maintenance)	88% (F1)	High	High	High	N/A
Logistic Regression (Churn)	78%	Moderate	Moderate	Moderate	0.70
Random Forests (Churn)	85%	High	High	High	0.85



Conclusion

Across all case studies, the effectiveness of the machine learning models depended on the nature of the data, the complexity of the relationships between features, and the requirements of the specific industry.

Non-linear models such as Multilayer Perceptron (MLP) and Random Forests consistently outperformed simpler linear models in terms of predictive power, especially in disease diagnosis and customer churn prediction where complex interactions exist between features.

Interpretability vs. Accuracy: While models like SVM and MLP excel in predictive performance, their complexity can be a drawback in industries where interpretability is crucial. In such cases, simpler models like Decision Trees and Logistic Regression offer easier interpretation at the cost of reduced accuracy.

Imbalanced Data Handling: In domains like predictive maintenance and customer churn prediction, where data is often imbalanced, models that focus on metrics such as AUC-ROC, precision, and recall (e.g., Random Forests and SVM) were more effective in balancing the trade-off between false positives and false negatives.

Classification

From: Artificial intelligence in disease diagnostics: A critical review and classification on the current state of research guiding future direction

Type	Algorithm	Description
Supervised learning	Decision Tree	Decision trees classify instances through sorting them based on their individual value [155]. A decision tree always consists of a root node and several inner nodes and at least two leaves. Each node represents a logical rule and each leaf an answer to the problem
	Logistic Regression	Logistic regression algorithms aim at finding relationships between variables, which are refined in multiple iterations to predict an output value (or multiple values) based on given input features [156]. The output might be a finite number of states
	Naive Bayes	Naive Bayes is based on the Bayes theorem and represents a simple graphical model (directed acyclic graph) for the determination of relationships among various features [155]. The goal is to present the most compact probability distribution of involved variables by using known conditional independence
	Nearest Neighbor	Nearest neighbor is used for assessing the probability of classification (class membership) or regression (property value) [157]. The outcome is based on the nearest similar values (e.g., nearest neighbors)
	Neural Network	Neural networks are networks based on connected artificial neurons comparable to the human brain. Information flows through different layers within the network, whereby the final result can be manipulated by weighting the individual neurons [155]
	Random Forest	Random forest algorithms are a type of decision trees building up a multitude of trees with ranging depths [158]
	Support Vector Machine	Support vector machines classify subjects into two groups using a hyperplane (support vector), i.e. a two-dimensional space [33, 36]
Deep learning	Convolutional Neural Network	Convolutional neural networks are based on neural networks but are useful for processing data with multiple spaces, for example 2D- or 3D-images [61], by using an activation function for each neuron
	Deep Neural Network	Deep neural networks are highly complex neural networks with substantially higher amounts of layers [159]
	Recurrent Neural Network	Recurrent neural network is – like convolutional neural network, useful for but data with multiple space [61] but uses output of one neuron used as input for another or itself feedback loops, i.e., the output of a neuron suits as input for another neuron or even itself
Other	Others	There are other, less-known approaches, such as dimensionality reduction or regularization algorithms. However, in the context of this study, these play an insignificant role, which is why no detailed description is provided at this point

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Datasets

1. Dataset 1
2. Dataset 2
3. Dataset 3